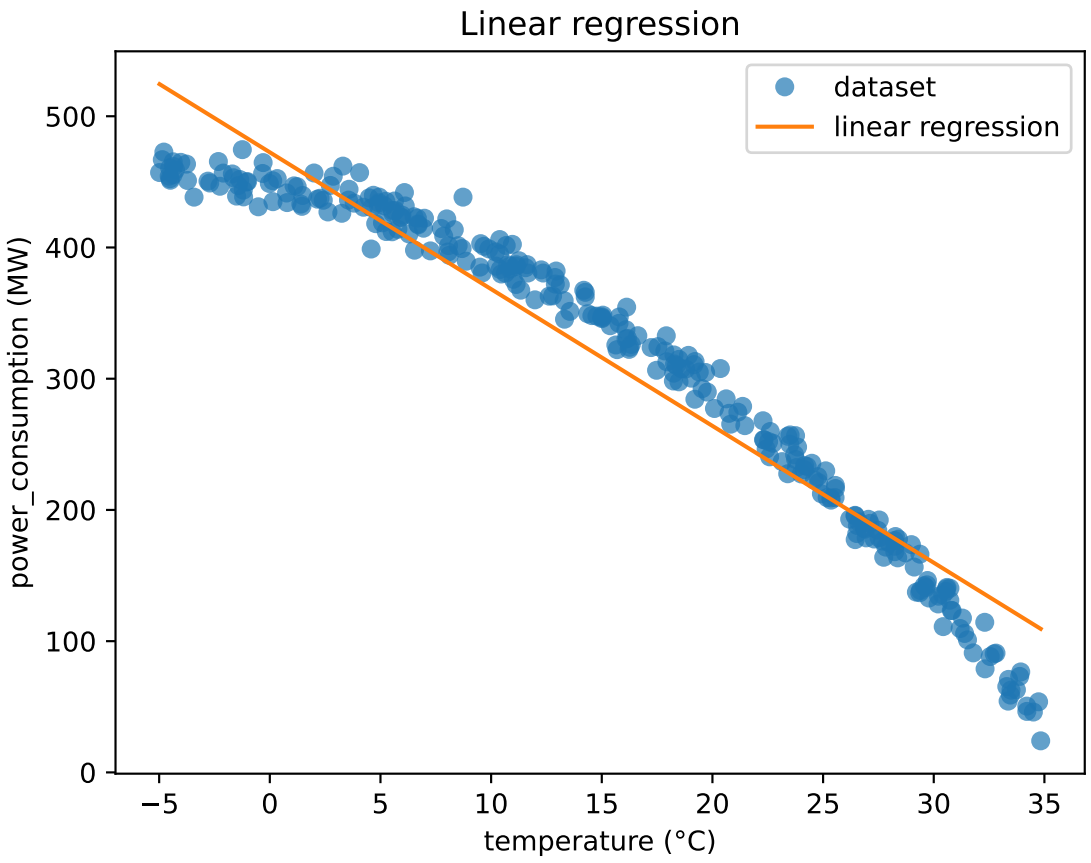
Fondamentaux théoriques du machine learning



Overview of lecture 2

Mathematical toolbox for ML (part I)

- Linear algebra
- Metrics
- Statistics, probability theory

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- Linear algebra
- Metrics
- Statistics, probability theory

Mathematical toolbox

- The aim of the course if to give an introduction to **fundamental principles** in ML.
- To do so, we will need an adapted mathematical toolbox and a bag of important results.

Why are mathematical aspects useful ?

- they allow a good comprehension of some theoretical results on ML
- these results allow a good choice of algorithms on practical problems (hopefully fast, accurate, etc.)

This section will give you an overview of the tools that will make you benefit more from the course if you are comfortable with them.

Matricial calculus

In machine learning, optimization or statistics we often write the inner product of two vectors of $\mathbb{R}^d$ as a product of matrices. If $x \in \mathbb{R}^d$ writes :

$$x = \begin{pmatrix} x_1 \\ \dots \\ x_i \\ \dots \\ x_d \end{pmatrix}$$

And (with T denoting the transposition),

$$y^T = \left(y_1, \dots y_j, \dots, y_d\right)$$

Then we have that

$$\langle x, y \rangle = y^T x = x^T y$$

Metrics

Let $D = \{x_1, \ldots, x_n\} \subset \mathcal{X}$ be a dataset of n samples, with labels $\{y_1, \ldots, y_n\} \subset \mathcal{Y}$.
There is a metric in the input space $\mathcal{X}$ and in the output space $\mathcal{Y}$.

- The **metric** in $\mathcal{X}$ determines to what extent two samples x_i and x_j should be considered similar or dissimilar.
- The **metric** in $\mathcal{Y}$ determines to what extent two labels y_i and y_j should be considered similar or dissimilar.

This is very important during the complete processing of the data.

Metrics in output space

A **loss function** l is a map that measures the discrepancy between to elements of a set (for instance of a linear space).

$$l : \begin{cases} \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}_+ \\ (y, z) \mapsto l(y, z) \end{cases}$$

Typically, z can represent our prediction for a given input x, $z = \tilde{f}(x)$, and y the correct label.

Most common loss functions

- "0-1" loss for **binary classification** : $\mathcal{Y} = \{0, 1\}$ or $\mathcal{Y} = \{-1, 1\}$.

$$l(y, z) = 1_{y \neq z} \tag{1}$$

- square loss for **regression** : $\mathcal{Y} = \mathbb{R}$.

$$l(y, z) = (y - z)^2 \tag{2}$$

- absolute loss for **regression** : $\mathcal{Y} = \mathbb{R}$.

$$l(y, z) = |y - z| \tag{3}$$

In **unsupervised learning**, there is notion of **output space !** (this might depend on the point of view)

Geometric distances in $\mathbb{R}^p$

$x = (x_1, ..., x_p)$ and $y = (y_1, ..., y_p)$ are p-dimensional **vectors**.

- $L_2 : ||x - y||_2 = \sqrt{\sum_{k=1}^{p}(x_k - y_k)^2}$ (Euclidian distance, 2-norm distance)
- $L_1 : ||x - y||_1 = \sum_{k=1}^{p} |x_k - y_k|$ (Manhattan distance, 1-norm distance)
- weighted $L_1 : \sum_{k=1}^{p} w_k |x_k - y_k|$, wich each $w_k > 0$
- $L_\infty : \max(x_1, \ldots, x_n)$ (infinity norm distance, Chebyshev distance)

Choice of the metric

In some contexts, some usual metrics such as $L2$ might not be meaningful !

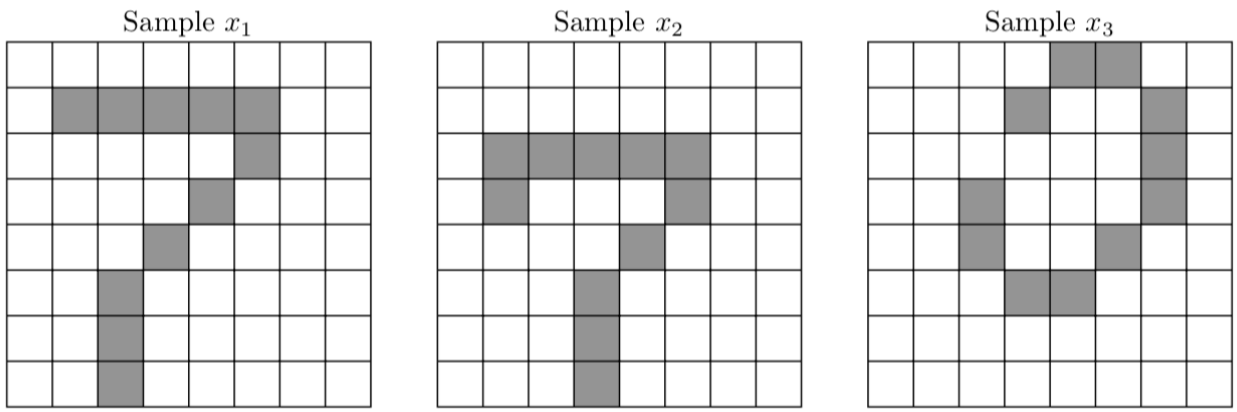
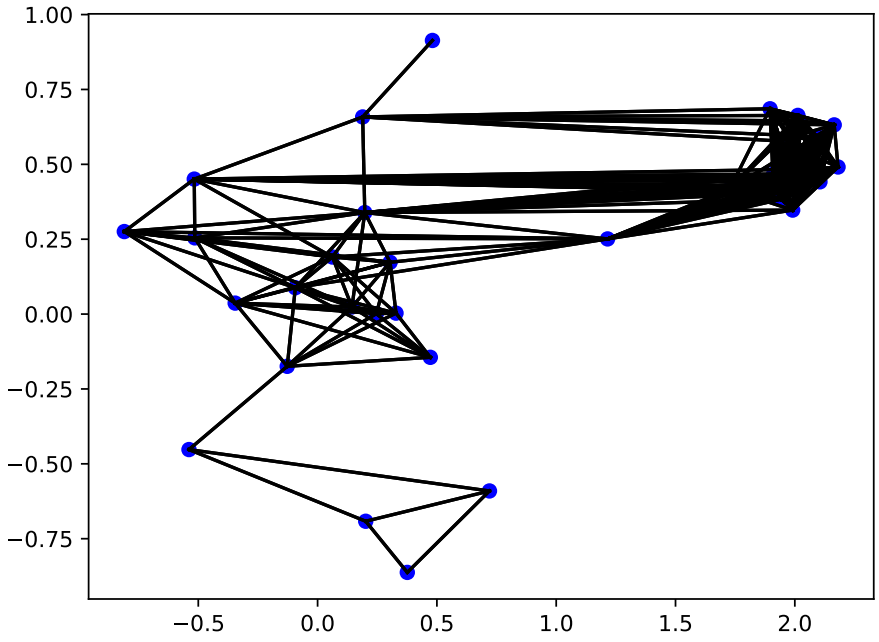
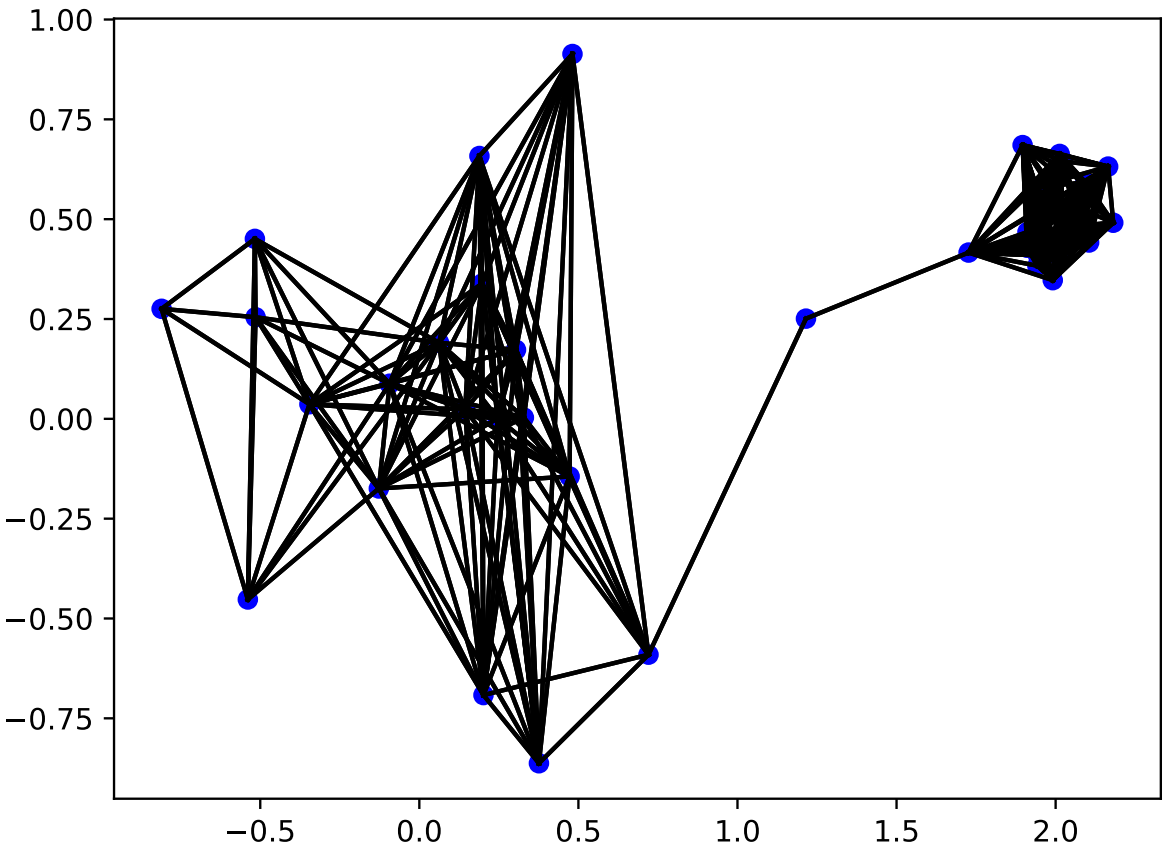


Figure – In $\mathbb{R}^{64}$, those three points form an equilateral triangle, [Fix et al., ,]

Exercice 1 : Using **simulations/lecture_2/metrics/geometric_data/build_graphs.ipynb**, choose the metric and the threshold so that this graph (and the ones on the next slides) are built.

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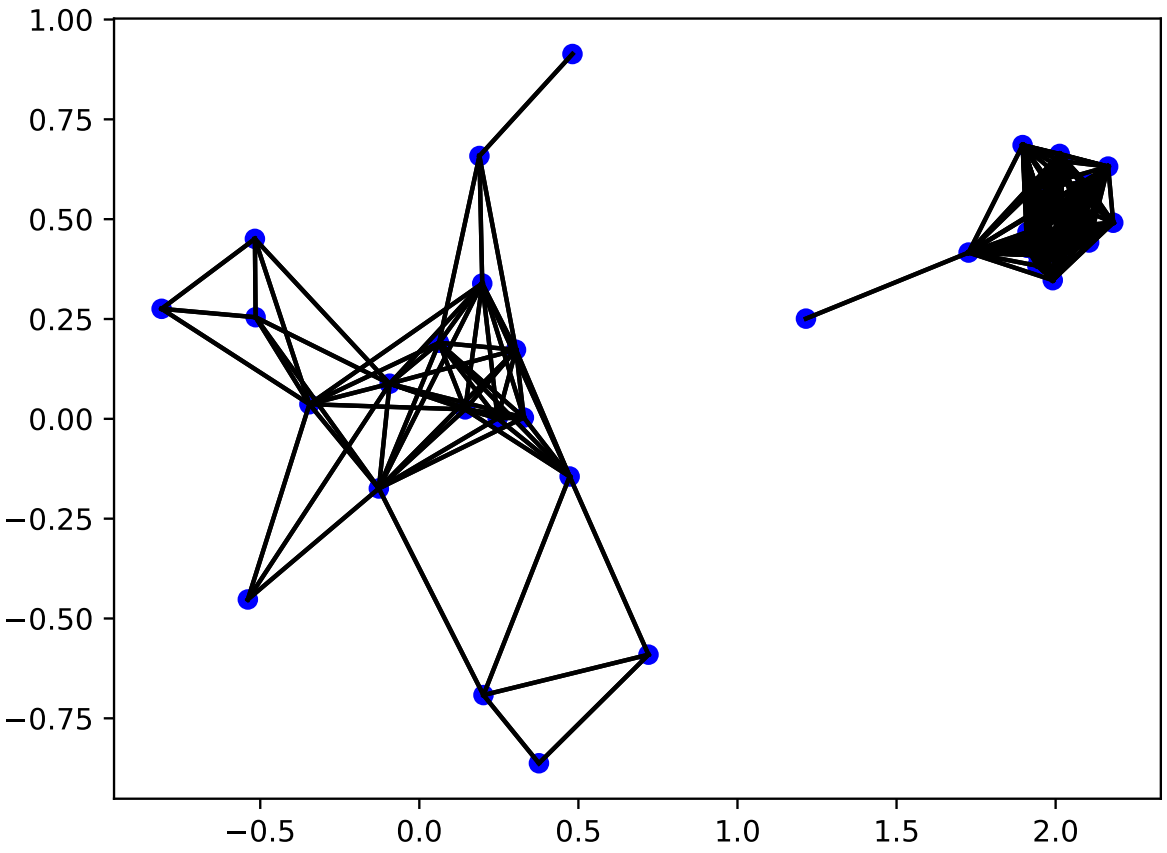
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Non-geometric data

Not all data are geometric !

Hamming distance

- $\#\{x_i \neq y_i\}$ (Hamming distance)
- Levenshtein distance for strings (allows deletions and additions)

General definition of a distance

A **distance** on a set E is an application $d : E \times E \to \mathbb{R}_+$ that must :

- be **symetric** : $\forall x, y, d(x, y) = d(y, x)$
- **separate the values** : $\forall x, y, d(x, y) = 0 \Leftrightarrow x = y$
- respect the **triangular inequality** $\forall x, y, z, d(x, y) \leq d(x, z) + d(y, z)$

Similarities

Sometimes, it is not possible to define a proper **distance** in the input space $\mathcal{X}$! This may happen for instance is $\mathcal{X}$ is a dataset of texts.

- When distances are unavailable, we can use **Similarities** or **Dissimilarity** to compare points.
- Dissimilarites are more general and don't always abide by the distance axioms.
- Other examples : Adjacency in an oriented graph, custom agregated score to compare data.

Example : cosine similarity

The **cosine similarity** may be used to compare texts.
If u and v are vectors,

$$S_C(u, v) = \frac{(u|v)}{||u||||v||} \tag{4}$$

- the **bag of words representation** allows us to build a vector from a text (one hot encoding).
- **cosine_similarity/scraper.py**
- **cosine_similarity/similarity.py**

Hybrid data

Sometimes each sample contains both numerical data and non-numerical data (text, categorical data.)
See **hybrid_data/**
This is often the case in machine learning applications! (database of customers, database of cars, etc.)

Moments of a distribution

Definition

Moments of a distribution

Let X be a real random variabe on a probability space Ω, and $k \in \mathbb{N}^*$. X is said to have a moment of order k if $E(|X|^k) < +\infty$, which means that :

- if X is discrete, with image $X(\Omega) = (x_i)_{i \in \mathbb{N}}$, the series

$$\sum (x_i)^k P(X = x_i)$$

  is **absolutely** convergent. The moment is then equal to the sum of that series (without absolute value).

Moments of a distribution

Definition

Moments of a distribution

Let X be a real random variabe on a probability space Ω, and $k \in \mathbb{N}^*$. X is said to have a moment of order k if $E(|X|^k) < +\infty$, which means that :

- if X is continuous with density $p(x)$, the integral

$$\int_{-\infty}^{+\infty} x^k f(x) dx$$

  is **absolutely** convergent. The moment is then equal to the integral (without absolute value).

Moments of a distribution

Proposition

Let $k_1 < k_2$ be integers. Let X be a real random variable. Then if X has a moment of order k_2, X also has a moment of order k_1.

Moments of a distribution

Exercice 2 : **Prove the proposition**

Proposition

Let $k_1 < k_2$ be integers. Let X be a real random variable. Then if X has a moment of order k_2, X also has a moment of order k_1.

Expected value, variance

Definition

Expected value, variance

- If X has a moment of order 1, it is called the **expected value**
- If X has a moment of order 2, then $X - E(X)$ also has a moment of order 2. This moment is called the variance of X.

$$V(X) = E\big((X - E(X))^2\big)$$

We often note $\sigma(X) = \sqrt{Var(X)}$ (standard deviation).

Expected value, variance

Proposition

Let a and b be real numbers, and X a random variable that admits a moment of order 2. Then

$$Var(aX + b) = a^2 Var(X)$$

Independence

Proposition

Let $(X_1, \ldots, X_n)$ *be* n *mutually independent real random variables. Then if they all admit a moment of order* 1*, then the product* $X_1 X_2 \ldots X_n$ *also does admit a moment of order* 1 *and*

$$E(X_1 X_2 \ldots X_n) = \prod_{i=1}^{n} E(X_i)$$

If they also admit moments of order 2*, then*

$$Var(\sum_{i=1}^{n} X_i) = \sum_{i=1}^{n} Var(X_i)$$

Covariance

The covariance of the two real variables X and Y writes

$$Cov(X, Y) = E[(X - E[X])(Y - E[Y])] \tag{5}$$

Lemma

Let $X, Y, Z \in \mathbb{R}$ be real random variables with a moment of order 2. We have :

$$Cov(X + Y, Z) = Cov(X, Z) + Cov(Y, Z)$$

$$Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$$

$$|Cov(X, Y)| \leq \sigma(X)\sigma(Y)$$

Convention

From now on, if we write $E(X)$ or $Var(X)$, we implicitely assume that the quantities are correctly defined.

Random vectors

Definition

Let $X \in \mathbb{R}^d$ be a random vector.

$$X = \begin{pmatrix} X_1 \\ \dots \\ X_i \\ \dots \\ X_d \end{pmatrix}$$

The **expected value** of the vector writes

$$E(X) = \begin{pmatrix} E[X_1] \\ \dots \\ E[X_i] \\ \dots \\ E[X_d] \end{pmatrix}$$

Random vectors

Definition

$$X = \begin{pmatrix} X_1 \\ \dots \\ X_i \\ \dots \\ X_d \end{pmatrix}$$

The **variance matrix** (or **covariance matrix, variance-covariance, dispersion matrix**) $Var(X)$ is defined as

$$[Var(X)]_{ij} = Cov(X_i, X_j)$$

Random vector

Exercice 3 : **Random vector**

What does it mean to have a vector such that

$$Var(X) = \lambda I_d \tag{6}$$

?

Expected value as a minimization

Exercice 4 : **Expected value as minimization.**
Show that $E(X)$ is the value that minimizes the function

$$f(t) = E\big((X - t)^2\big) \tag{7}$$

Markov inequality

Proposition

Markov inequality
Let X be a real non-negative random variable (variable aléatoire réelle positive), such that $E(|X|) < +\infty$. Let $a > 0$. Then

$$P(X \geq a) \leq \frac{E(X)}{a}$$

Chebychev inequality

Proposition

Chebyshev inequality Let X ba a real random variable, such that $E(|X|^2) < +\infty$. Let $a > 0$. Then

$$P(|X - E[X]| > a) \leq \frac{Var(X)}{a^2}$$

Weak law of large numbers

Theorem

Weak law of large numbers

Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of i.i.d. variables that have a moment of order 2. We note m their expected value. Then

$$\forall \epsilon > 0, \lim_{n\to+\infty} P\big(|\frac{1}{n}\sum_{i=1}^{n} X_i - m| \geq \epsilon\big) = 0$$

*We say that we have **convergence in probability**.*

Standard deviation of the average

If $E(S_n) = m$, then

$$\sqrt{Var\left(S_n - m\right)} = \frac{\sigma}{\sqrt{n}} \tag{8}$$

References I

Fix, J., Frezza-Buet, H., Geist, M., and Pennerath, F.
Machine Learning.pdf.